

A single-cell, spatial transcriptomic atlas of the *Arabidopsis* life cycle

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Arabidopsis has been pivotal in uncovering fundamental principles of plant biology, yet a comprehensive, high-resolution understanding of its cellular identities throughout the entire life cycle remains incomplete. Here we present a single-nucleus and spatial transcriptomic atlas spanning ten developmental stages, encompassing over 400,000 nuclei from all organ systems and tissues—from seeds to developing siliques. Leveraging paired single-nucleus and spatial transcriptomic datasets, we annotate 75% of identified cell clusters, revealing striking molecular diversity in cell types and states across development. Our integrated approach identified conserved transcriptional signatures among recurrent cell types, organ-specific heterogeneity and previously uncharacterized cell-type-specific markers validated spatially. Moreover, we uncover dynamic transcriptional programs governing secondary metabolite production and differential growth patterns, exemplified by detailed spatial profiling of the compact yet complex apical hook structure; this profiling revealed transient cellular states linked to developmental progression and hormonal regulation, highlighting the hidden complexity underlying plant morphogenesis. Functional validation of genes uniquely expressed within specific cell contexts confirmed their essential developmental roles, underscoring the predictive power of our atlas. Collectively, this comprehensive resource provides an invaluable foundation for exploring cellular differentiation, environmental responses and genetic perturbations at high resolution, advancing our understanding of plant biology.

Multicellular organisms have evolved distinct organs to perform specific complex tasks and address challenges that may be present throughout their life. Unlike most animals, plants continuously undergo post-embryonic organogenesis to form new organs throughout development. In the model plant *Arabidopsis*, many tissues and organs present throughout its life cycle are composed of identical or similarly classified cell types that share functionality across organs and developmental stages¹. For example, epidermal cells are the interface of the plant with the external environment for all organs, while internally

localized vascular cells universally function in water and nutrient transport^{2,3}. Despite the current understanding of plant anatomy, there is limited understanding of the molecular constituents that define cell types and states within the native context of diverse organs present throughout development.

While cell-type marker genes exist for many recurrent cell types across distinct organs, the form and function of cell types can vary depending on the tissue context or cellular niche from which they originate. Intra-cell-type heterogeneity is exemplified by *Arabidopsis* sepal

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